From_Brain_to_Behavior slide 2

Brain activity produces perception and action

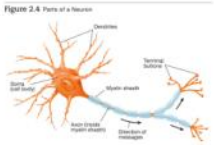
Behavioral neuroscience studies how the nervous system produces perception, movement, memory, and emotion. A **neuron** is a nerve cell. Neurons receive signals, change their electrical activity, and send signals to other cells. Incoming signals can increase or decrease a cell's tendency to fire.

Sensory systems detect events outside and inside the body. Motor systems organize actions such as reaching, walking, or speaking. Memory allows an earlier experience to change a later response. For example, seeing food can guide the direction of a reach. Hunger changes whether the person starts that action, and a remembered location changes where the person looks first.

An experiment can measure both the action and the activity of the cells involved. The scientist asks what the animal does, which cells are active, and how changing those cells changes the action. Recording identifies when activity occurs during the action. Changing the activity tests whether the action also changes, giving a more specific explanation than describing the movement alone.



Carlson and Birkett (2022), Fig. 1.5, p. 21. Cajal's drawing of Golgi-stained neurons.



Carlson and Birkett (2022), Fig. 2.4, p. 40. Labeled cell body, dendrites, axon, and terminals.

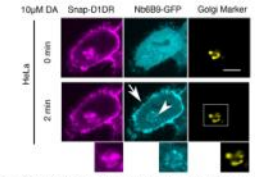
Chemical_Control slide 10

The same receptor class operates in different compartments

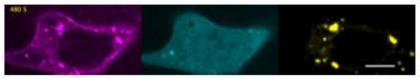
Externally applied dopamine activated surface and Golgi D1 receptors in HeLa cells. In HEK293 cells, a second laboratory cell line, it activated the surface receptors but produced little Golgi activation. The membrane-permeant agonist SKF81297 activated Golgi receptors in both cell types. The internal receptor population remained responsive even in cells where dopamine failed to reach it effectively.

HeLa cells expressed substantially more OCT2 than HEK293 cells. Introducing additional OCT2 into HEK293 cells allowed dopamine to activate their Golgi D1 receptors. Dopamine's chemical structure had not changed. The added transporter supplied a membrane-crossing route that the original cells lacked at a level sufficient to generate the internal response.

Golgi activation also persisted when the researchers blocked **endocytosis**, the process that moves surface membrane and its proteins into the cell. D1 receptors already located at the Golgi could respond after dopamine arrived. The fluorescent signal inside the cell therefore did not require activated surface receptors to travel inward during the experiment.



Puri et al. (2022), Fig. 1 (b, HeLa). D1 receptor activation at two cellular locations.



Puri et al. (2022), fig1-video2. Dopamine applied to D1 receptor-expressing HEK293 cells.

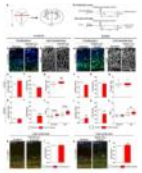
Chapter_16_slide 10

Overgrowth changes both cell number and cellular size

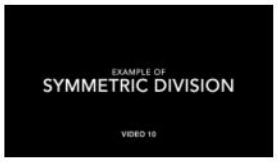
The Pik3ca models allowed researchers to examine why the brain became enlarged. Increased proliferation produced additional cells, and increased cell size expanded the tissue further. These are distinct cellular contributions to a similar visible anatomical outcome.

The cerebral cortex also developed abnormal organization. Cortical dysplasia means that the tissue's cellular arrangement is atypical, rather than simply larger. Such structural differences can change the routes through which excitatory and inhibitory neurons influence one another.

Normal neural stem cells can also increase cell number. Separate imaging of adult mouse hippocampus follows a stem cell dividing into two daughters that retain radial processes. These symmetric divisions are normal stem-cell behavior, unlike the abnormal cortical growth in the mutant mice.



Roy et al. (2015), Fig. 3.



Symmetric neural stem-cell divisions. Guo et al. (2022).

Play video

Chapter_18_slide 2

Stress coordinates the body with a challenge

A stressor is a demand that threatens stability or requires adaptation. Stress responses change circulation, metabolism, attention, and behavior so that an animal can respond. Their usefulness depends on the challenge, how long it lasts, and whether the animal can return to ordinary activity.

The hypothalamus helps organize these responses through autonomic nerves and hormone release. Its paraventricular nucleus contains neurons that produce corticotropin-releasing hormone, or CRH. These cells participate in the hormonal stress response and also influence which actions follow a stressful event.

After acute stress, a control mouse alternates among exploration, grooming, and other behaviors. Grooming is a normal body-care action, not a diagnosis of anxiety. The order and duration of these actions provide observable measures of how the animal reorganizes behavior after the challenge ends.



Füzesi et al. (2016), Fig. 1.



Control mouse after acute stress. Füzesi et al. (2016).

Play video

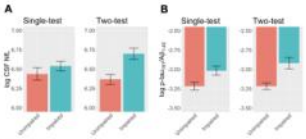
Football_Memory slide 10

A low score changes meaning across test definitions

Investigators compared two ways of defining cognitive impairment in former football players. One approach used a sufficiently low result on a single test. The other required lower performance on two tests within the same domain, such as memory or executive functioning, before classifying the domain as impaired.

The two-test definition identified fewer participants and showed associations with cerebrospinal-fluid markers of neurodegeneration that the single-test definition did not. An association with neurofilament light remained after excluding amyloid-positive participants. The paper examined how an assessment rule changes the biological characteristics of the group it identifies.

The separate examination recording shows a person who makes errors on orientation and memory questions. It demonstrates the behavior an examiner records, rather than the study's classification procedure. A visible error is an observation; deciding whether a cognitive domain is impaired requires the appropriate tests, norms, and clinical context.



Ly MT et al. (2025), Fig. 1.



Clinical demonstration: orientation errors. Larsen and Stensaas (2020).

Play video

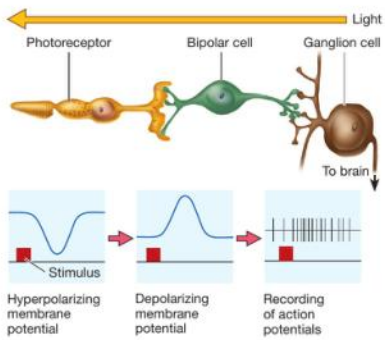
From_Light_to_Sight slide 10

Light reduces photoreceptor glutamate release

A **photopigment** contains an opsin protein bound to retinal, a light-sensitive molecule derived from vitamin A. Absorbing a photon changes retinal's shape and activates the opsin. In vertebrate rods and cones, the resulting chemical reactions reduce the concentration of cyclic GMP, a molecule that normally keeps certain membrane channels open.

As those channels close, inward positive current decreases and the photoreceptor **hyperpolarizes**, meaning its interior becomes more negative. This differs from a neuron becoming excited by opening an inward-current channel. The photoreceptor responds to increased light by reducing its electrical drive toward transmitter release at its synaptic ending.

In darkness, photoreceptors are relatively depolarized and release glutamate continuously. Light-induced hyperpolarization reduces calcium entry into the terminal and lowers glutamate release. Bipolar cells detect that decrease. Different bipolar-cell receptors then convert the same reduction in glutamate into either an increase or a decrease in their own electrical response.



Carlson and Birkett (2022), Fig. 6.8, p. 160.